Polynomial equations

1 Solve the following equations:

a $(5x-4)(x+3)(x-2)=0$

b $\left(x^2-16\right)\left(x^2+12x+36\right)=0$

c $(x+3)^2(x-6)(5x-2)=0$

d $3x^3=5x^2$

e $x^3-125=0$

f $x^3+5x^2-50x=0$

g $x^4-16x^2=0$

h $64x^4-81x^2=0$

2 Consider the equation $2x^5+36x^4-80x^3=0$.

a Rewrite the equation in factored form.

b Solve the equation.

3 Consider the equation $4x^3-12x^2-5x+15=0$.

a Rewrite the equation as the product of a linear factor and a quadratic factor.

b Solve the equation.

4 It is given that $x=5$ is a root of the equation $x^3+x^2-21x-45=0$.

a Use long division or otherwize to rewrite the equation as the product of a linear factor and a quadratic factor.

b Solve the equation for the other root(s).

Equations reducible to quadratics

5 For each of the following equations:

i Using the variable p, determine a substitution that will reduce the equation to a quadratic in terms of p.

ii Make the substitution, and solve the new equation for p.

iii Solve the original equation for x.

a $x^4-34x^2+225=0$

b $x^4-26x^2+25=0$

c $25x^4-65x^2+36=0$

d $81x^4-117x^2+36=0$

e $(2x+4)^2+10(2x+4)+16=0$

f $(3x+3)^2+21(3x+3)+108=0$

g $(4x+8)^2+16(4x+8)+48=0$

h $(4x-4)^2+2(4x-4)-63=0$

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6 Solve the following equations:

a $x^4 - 5x^2 + 4 = 0$

b $x^4 - 20x^2 + 64 = 0$

c $x^4 - 49x^2 + 600 = 0$

d $x^4 - 34x^2 + 225 = 0$

e $(4x+9)^2 + 13(4x+9) + 42 = 0$

f $(8x+5)^2 + 17(8x+5) + 70 = 0$

g $3(9x+10)^2 + 19(9x+10) + 20 = 0$

h $2(8x-3)^2 + 13(8x-3) + 15 = 0$

7 Solve for the real roots of the following equations:

a $x^4 = 8x^2 + 9$

b $x^4 = 3x^2 + 4$

c $x^4 = 12x^2 + 64$

d $x^4 = 24x^2 + 25$

e $(x^2 - 2)^2 - 25 = 0$

f $(x^2 - 7)^2 - 64 = 0$

g $(x^2 - 5)^2 - 81 = 0$

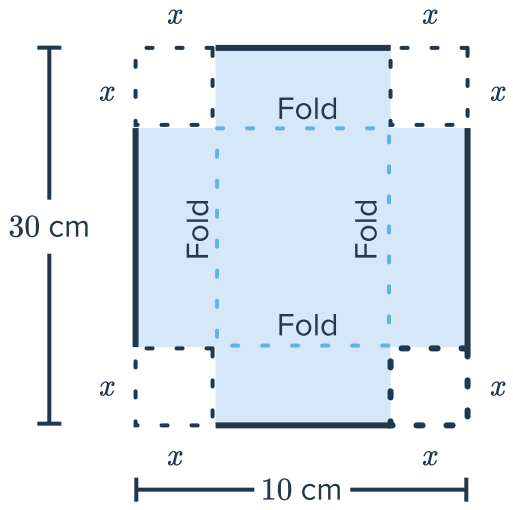
h $(x^2 + 6)^2 - 15 = 34$

Applications

8 A box is formed by cutting squares of length x cm from the corners of a piece of cardboard 10 cm by 30 cm.

a Find an expression for the volume of the box in terms of x.

b Find the value of x if the volume is 176 cm^3.



9 A retail outlet sells x units of a product each year. The annual cost for this product is given by $C = 9x + 3x^2$ and the annual gross profit is given by $G = 9 - 2x - 4x^2 + x^3$ where C and G are both in dollars.

a Form an expression for P, the net profit after producing and selling x units.

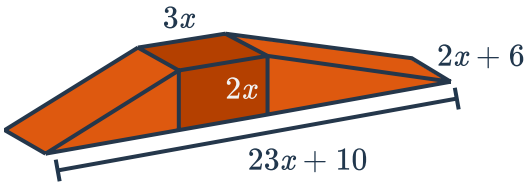
b Find the number of units that must be sold in order to make a profit of $\$40\,672$.

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10 A two-sided skate ramp is to have dimensions as shown in the image. One of the sloped sections is 50% longer (horizontally) than the other.

a If the total volume of the two sloped sections is $240\ \text{ft}^3$, construct an equation in terms of x to represent the situation.

b Determine the dimensions of the skate ramp.



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